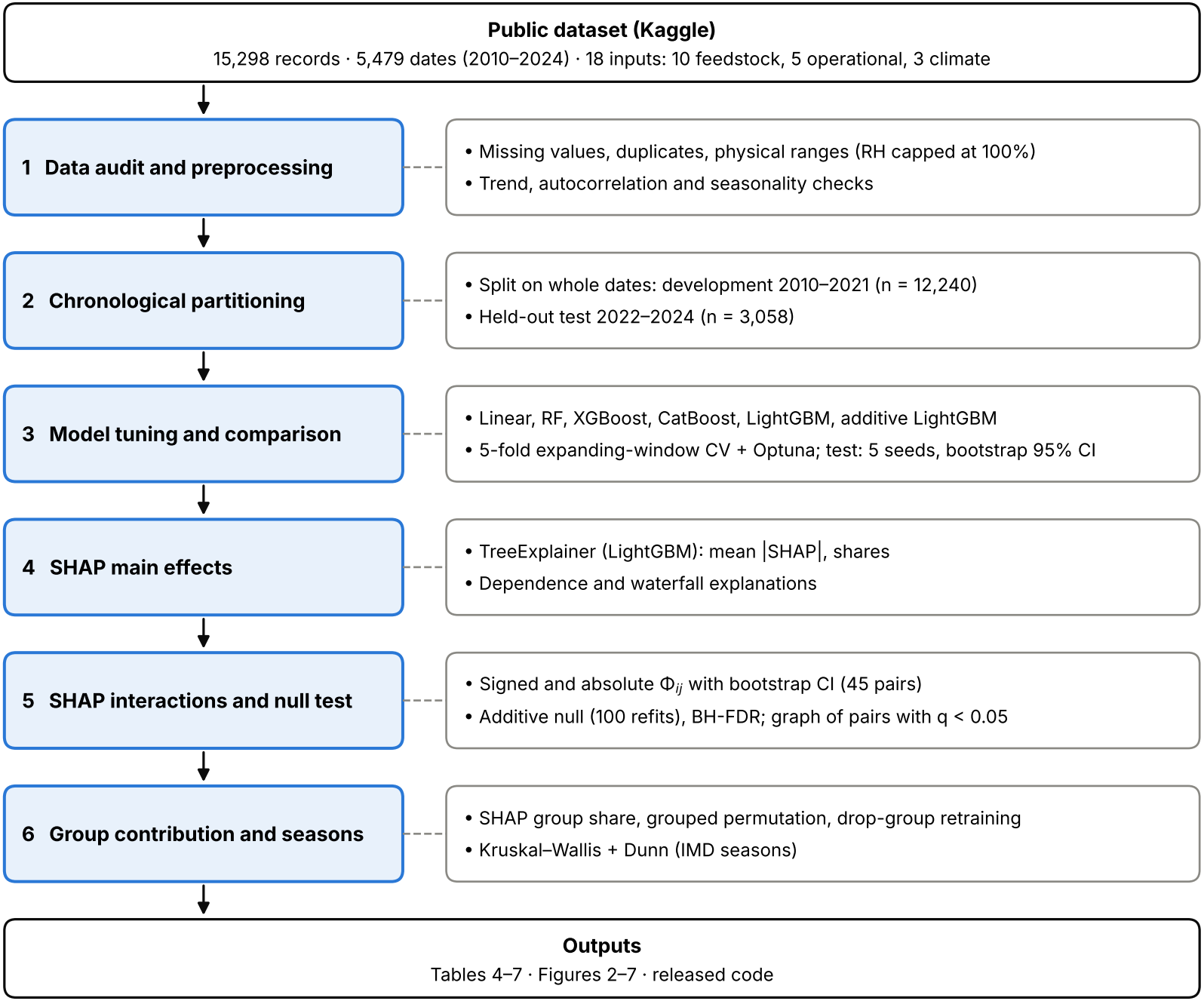
Public dataset (Kaggle)
15,298 records · 5,479 dates (2010–2024) · 18 inputs: 10 feedstock, 5 operational, 3 climate
1 Data audit and preprocessing
• Missing values, duplicates, physical ranges (RH capped at 100%)
• Trend, autocorrelation and seasonality checks
2 Chronological partitioning
• Split on whole dates: development 2010–2021 (n = 12,240)
• Held-out test 2022–2024 (n = 3,058)
3 Model tuning and comparison
• Linear, RF, XGBoost, CatBoost, LightGBM, additive LightGBM
• 5-fold expanding-window CV + Optuna; test: 5 seeds, bootstrap 95% CI
4 SHAP main effects
• TreeExplainer (LightGBM): mean |SHAP|, shares
• Dependence and waterfall explanations
5 SHAP interactions and null test
• Signed and absolute Φ_{ij} with bootstrap CI (45 pairs)
• Additive null (100 refits), BH-FDR; graph of pairs with q < 0.05
6 Group contribution and seasons
• SHAP group share, grouped permutation, drop-group retraining
• Kruskal–Wallis + Dunn (IMD seasons)
Outputs
Tables 4–7 · Figures 2–7 · released code